

MMF9 - Galingale View PMF Source Apportionment Analysis

Analysis Date: 2025-10-04 14:51

Data Period: 2023-10-01 to 2023-10-30

Factors Resolved: 6

Models Run: 1

[CONFIG] Run Configuration

- **Uncertainty Mode:** epa (EPA PMF 5.0 formulas)
- **ESAT Algorithm:** LS-NMF (Standard PMF, nonnegative)
- **Initialization:** Column Mean
- **Matrix Updates:** Standard training
- **Random Seed:** 42
- **Records Analyzed:** 7,984
- **Original Species:** 14
- **Final Species:** 14

[UNCERTAINTY MODE] EPA Uncertainty Policy (PMF 5.0)

- **Above MDL:** $U = \sqrt{(EF \times \text{conc})^2 + (0.5 \times \text{MDL})^2}$
- **BDL Cases:** $V = \text{MDL}/2$, $U = \text{Five Sixth Mdl}$
- **Missing Cases:** $V = \text{species median (fallback MDL)}$, $U = 4 \times \text{median (fallback } 4 \times \text{MDL)}$
- **EF/MDL Source:** Built-in database
- **Aggregation Scaling:** Applied as $1/\sqrt{n}$ when counts are available

[NUMBERS] S/N Categorization Summary (EPA PMF 5.0)

- **Total Species:** 14
- **Average S/N:** 3.028
- **Thresholds:** Strong ≥ 2.0 , Weak 0.2-2.0, Bad < 0.2

Category Breakdown:

- **Strong: 10 species** (normal weighting)
- **Weak: 4 species** (uncertainty × 3)
- **Bad: 0 species** (excluded from analysis)

Species Categories:

Species	Category	S/N	Action
Benzene	Weak	1.317	Uncertainty tripled
Ethylbenzene	Weak	0.706	Uncertainty tripled
H2S	Strong	2.233	Normal weighting
NO	Strong	3.581	Normal weighting
NO2	Strong	3.693	Normal weighting
NOX	Strong	3.876	Normal weighting
PM1 FIDAS	Strong	4.817	Normal weighting
PM10 FIDAS	Strong	3.907	Normal weighting
PM2.5 FIDAS	Strong	5.861	Normal weighting
PM4 FIDAS	Strong	4.737	Normal weighting
SO2	Strong	2.625	Normal weighting
TSP FIDAS	Strong	2.983	Normal weighting
Toluene	Weak	1.172	Uncertainty tripled
m&p-Xylene	Weak	0.881	Uncertainty tripled

[EXCLUSIONS] Species Exclusions

The following species were completely removed from PMF analysis via the --exclude-species flag:

Species	Reason
CH4	CLI exclusion flag

Note: Excluded species are completely removed from concentration and uncertainty matrices before PMF analysis.

[CLOSURE] Mass Closure / Fit Divergence Analysis

Purpose: Quantify how well the PMF model reconstructs measured concentrations (closure) and identify species where regularization affects fit quality.

Group-Level Closure:

- **Gases:** 100.0% (5 species)
- **VOCs:** 114.2% (4 species)
- **PM:** 99.3% (5 species)

Interpretation Guide:

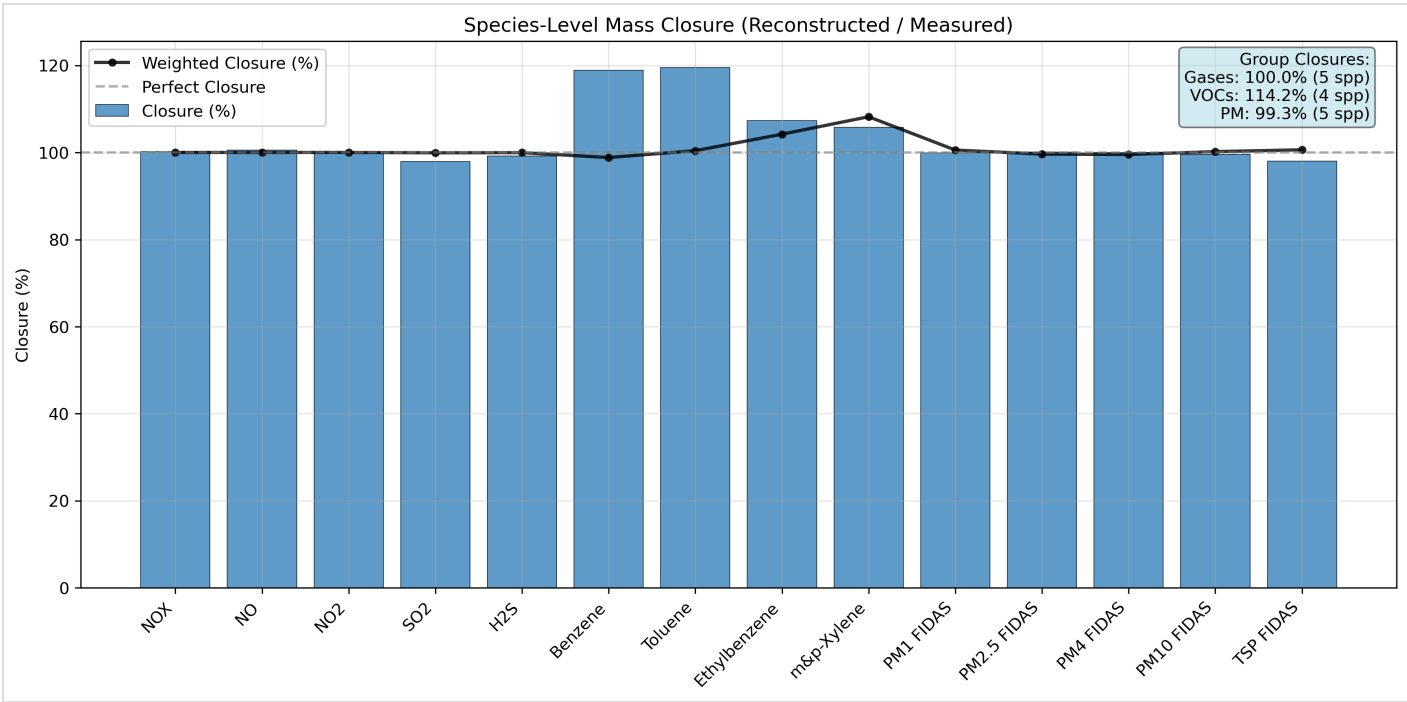
- **Closure %:** $(\text{Reconstructed Sum} / \text{Measured Sum}) \times 100$. Values near 100% indicate good fit.
- **Weighted Closure %:** Same calculation but weighted by $1/\text{uncertainty}^2$. More robust to outliers.
- **Red bars:** Regularized species (expected to show closure reduction as λ increases).
- **Q Share %:** Fraction of total model Q contributed by each species (indicates fit difficulty).

Files: See `MMF9_mmf_20231001_20231030_closure_summary.csv` for detailed per-species metrics.

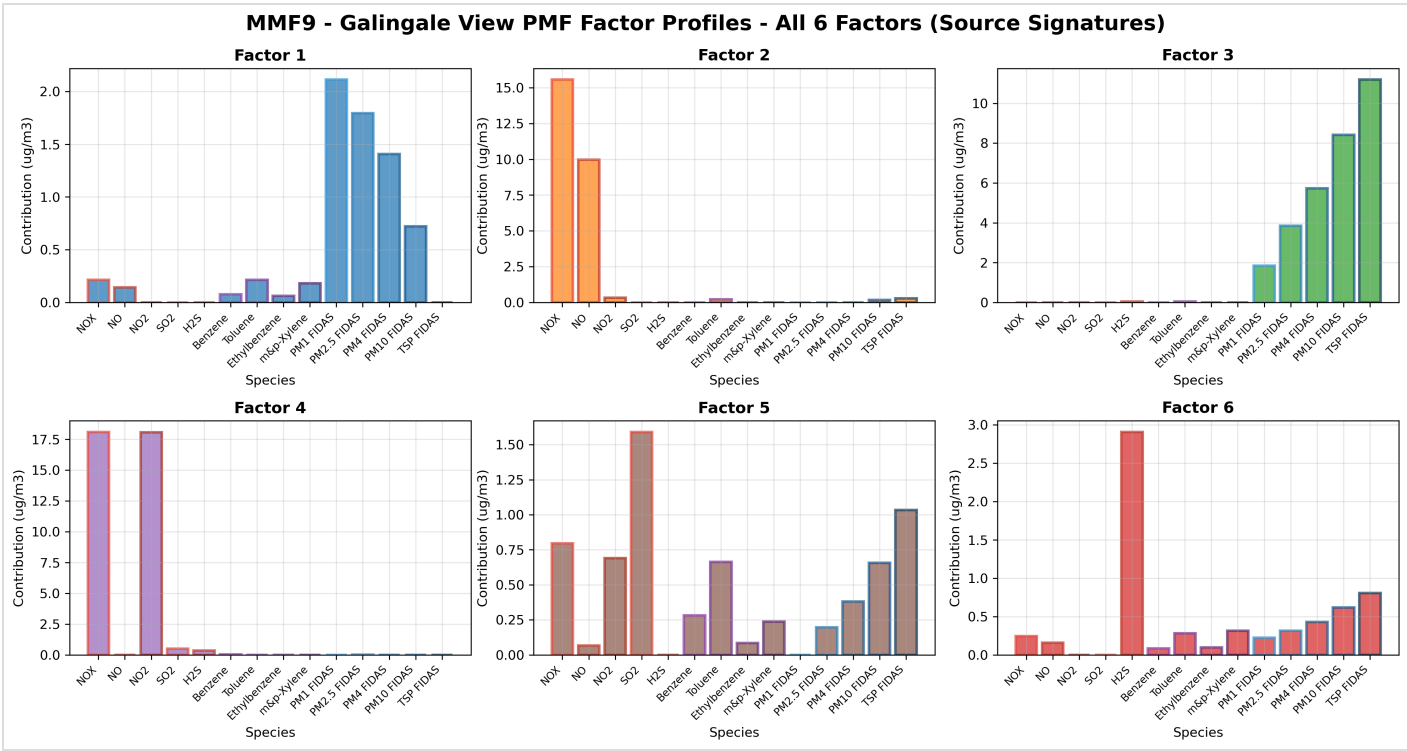
Model Performance

- **Q(true):** {self.best_model.Qtrue:.2f}
- **Q(robust):** {self.best_model.Qrobust:.2f}
- **DoF:** 63,788
- **Q/DoF Ratio:** 0.134 (Excellent per EPA guidelines)

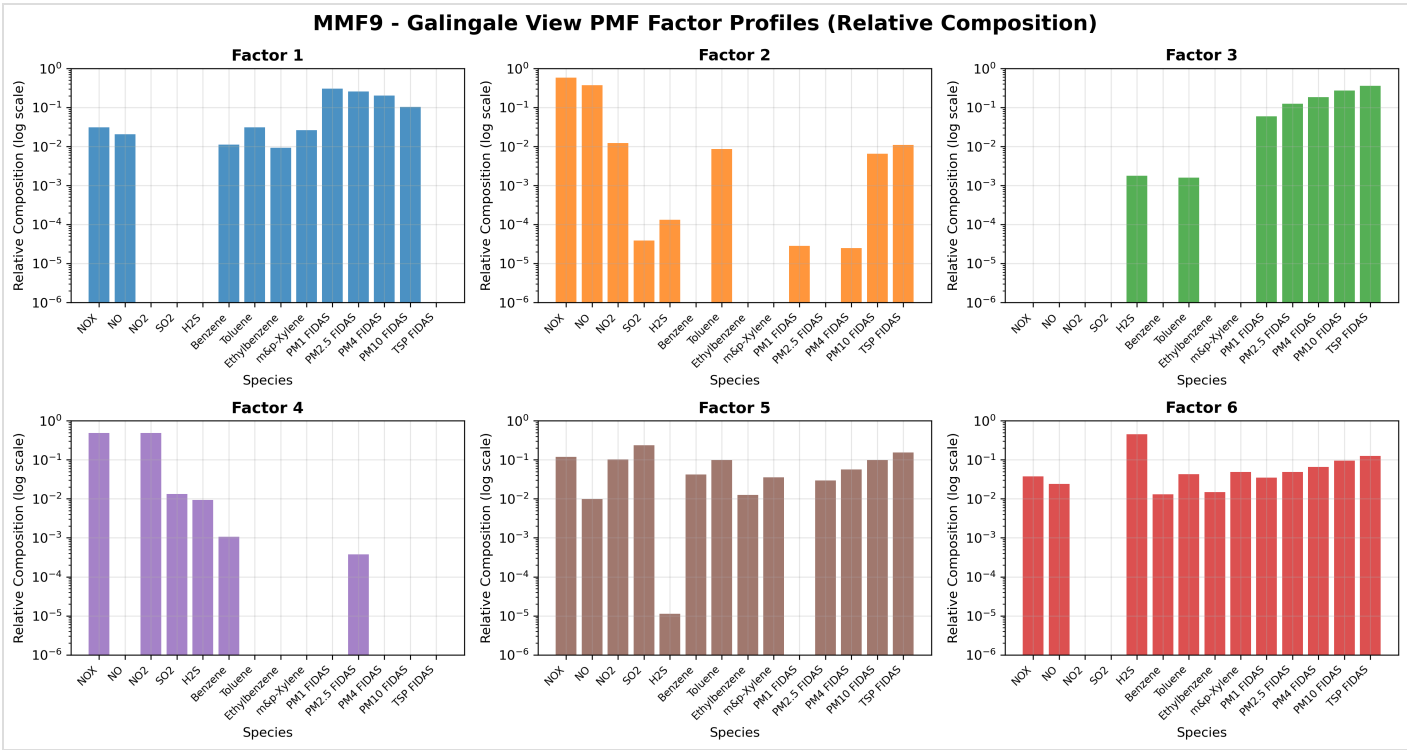
Mmf9 Mmf 20231001 20231030 Closure Summary



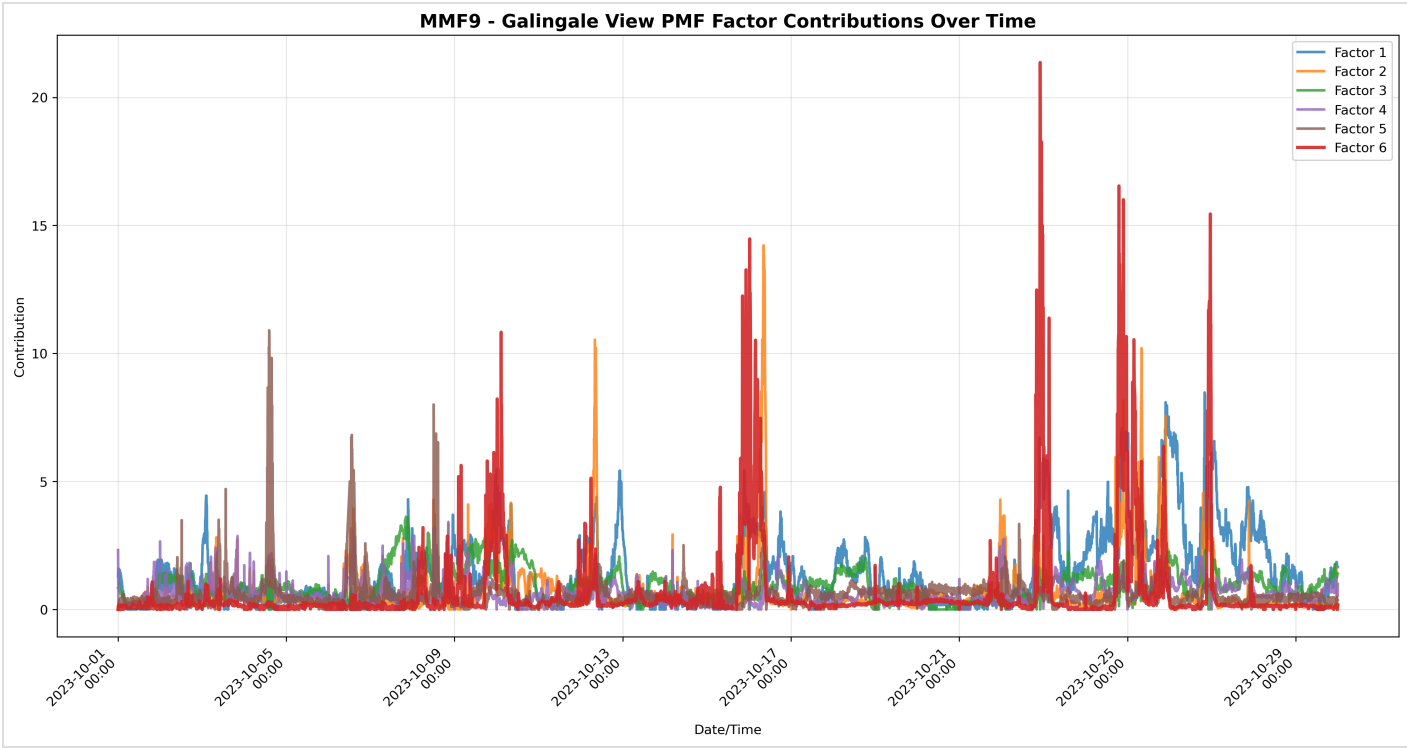
Mmf9 Mmf 20231001 20231030 Factor Profiles



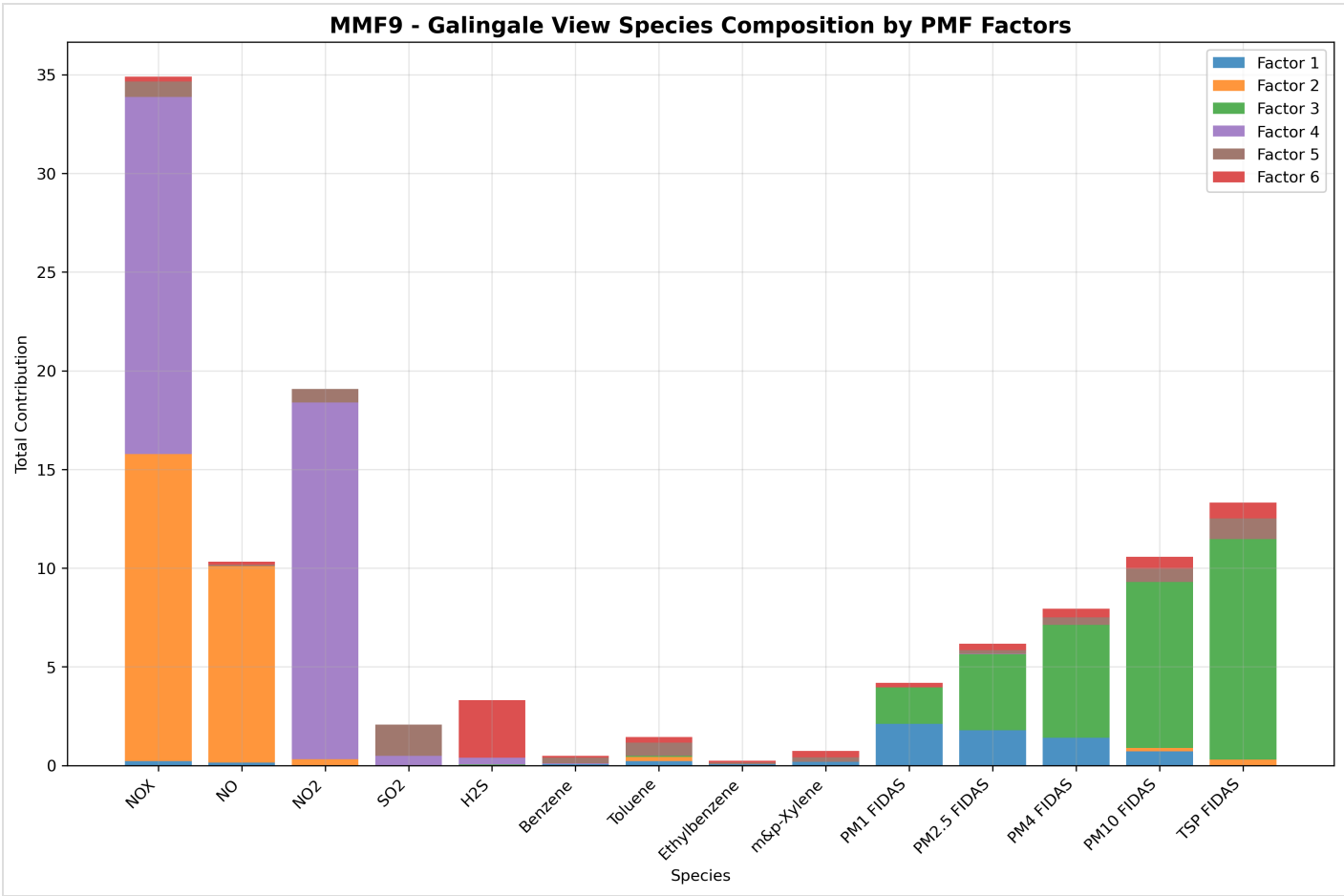
Mmf9 Mmf 20231001 20231030 Factor Profiles Relative



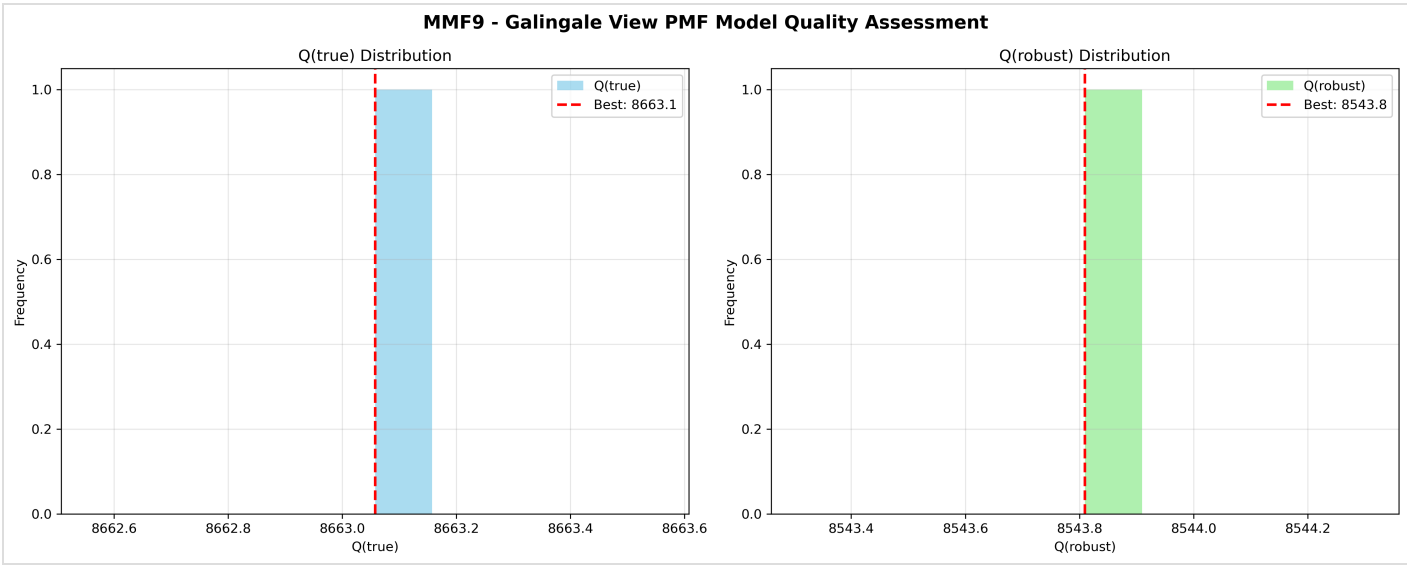
Mmf9 Mmf 20231001 20231030 Factor Contributions



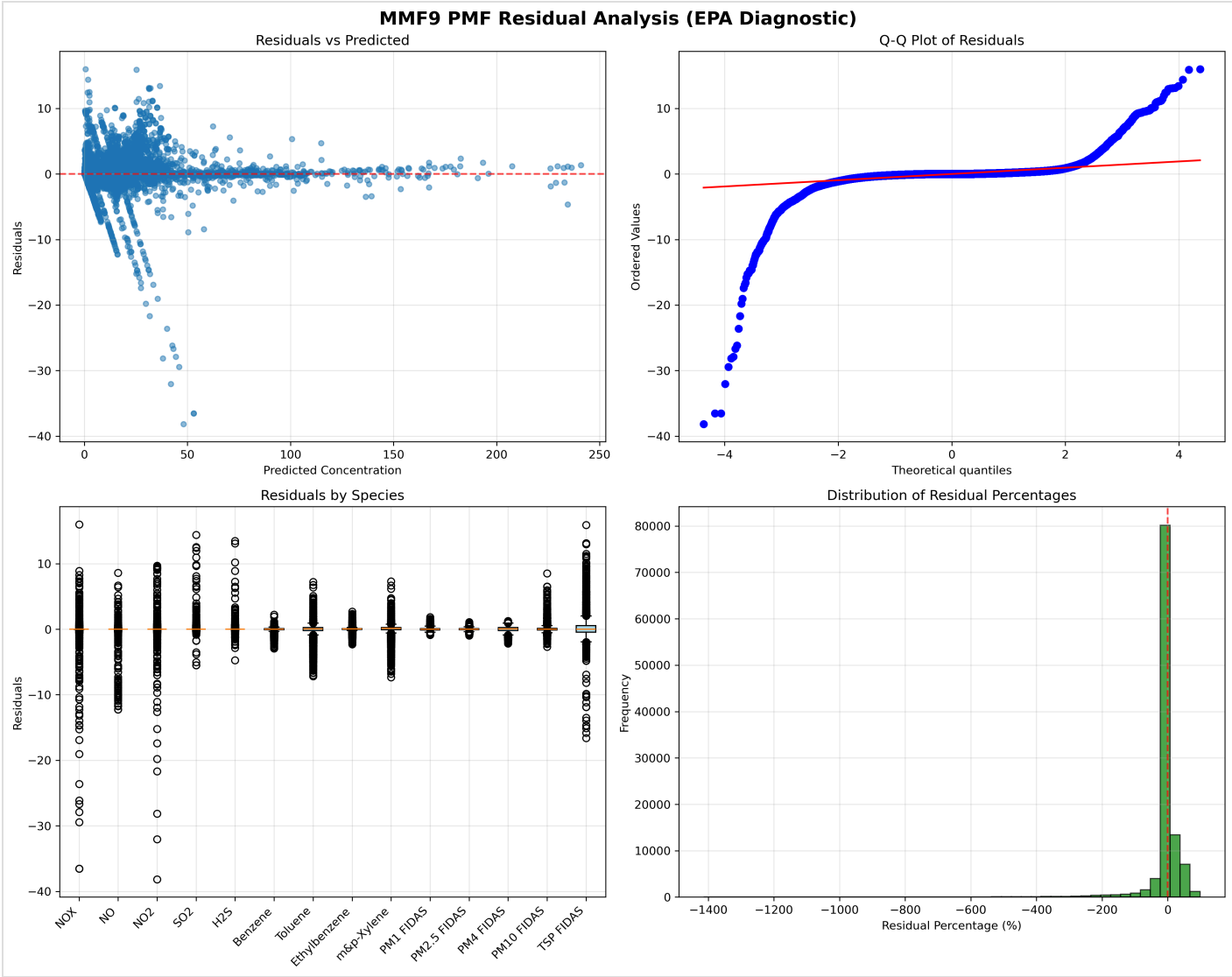
Mmf9 Mmf 20231001 20231030 Species Composition



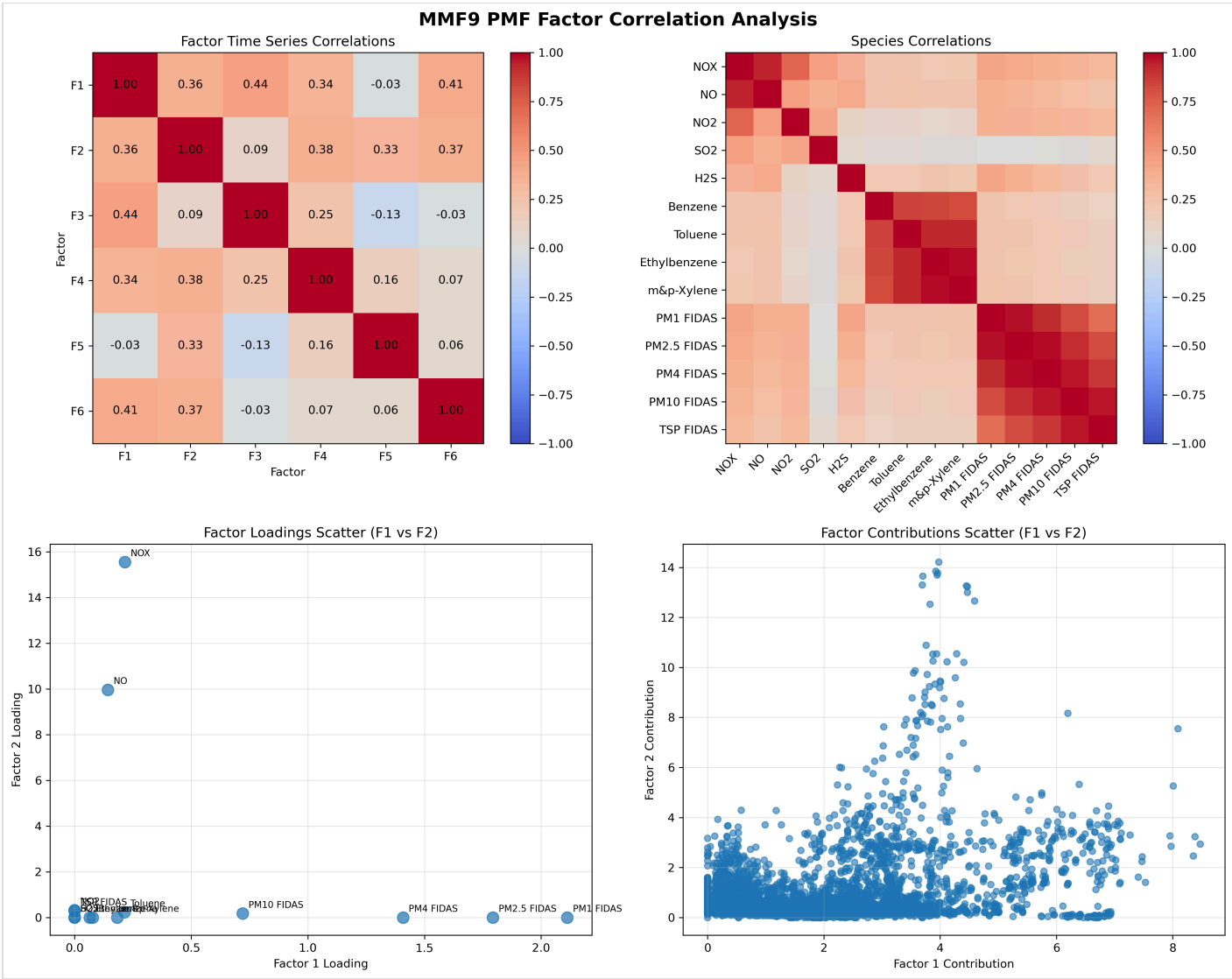
Mmf9 Mmf 20231001 20231030 Model Quality



Mmf9 Mmf 20231001 20231030 Residual Analysis



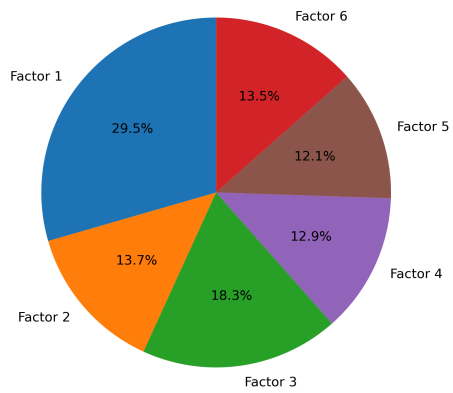
Mmf9 Mmf 20231001 20231030 Correlation Analysis



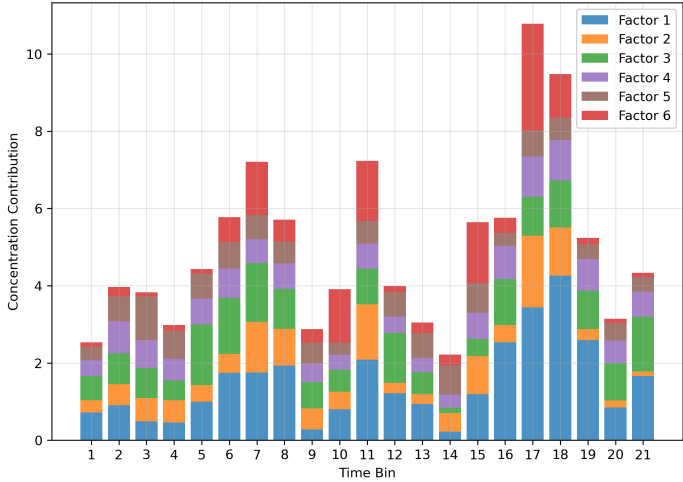
Mmf9 Mmf 20231001 20231030 Source Contribution Analysis

MMF9 PMF Source Contribution Analysis

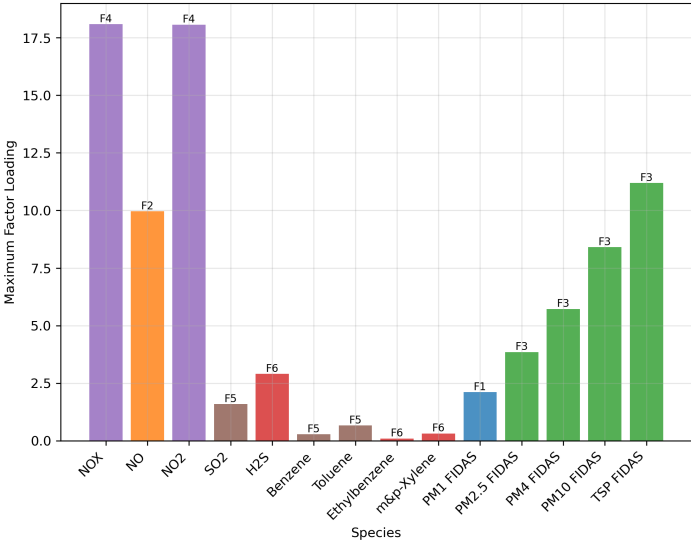
Relative Source Contributions



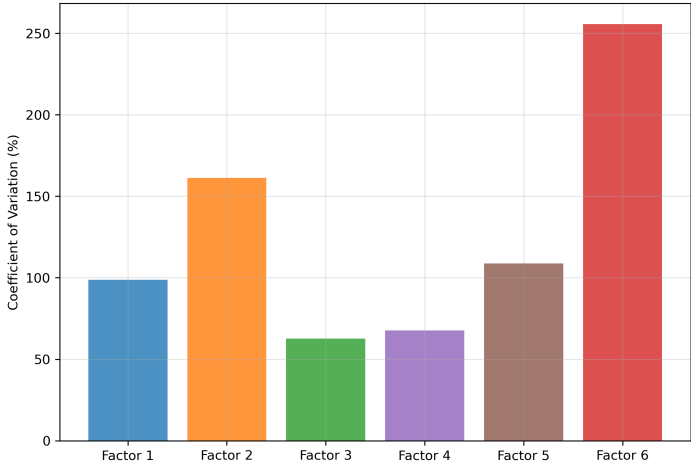
Source Contributions Over Time (Binned)



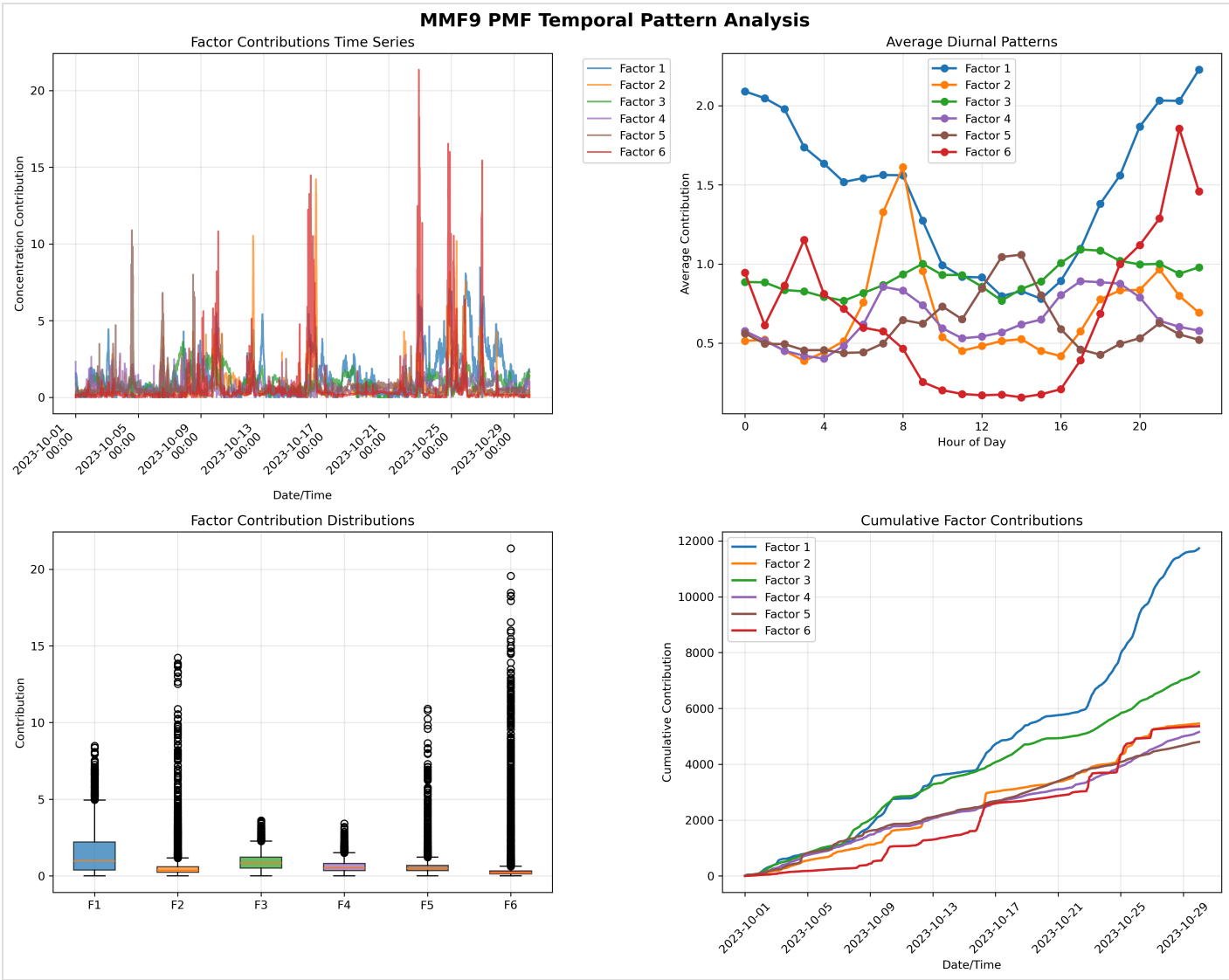
Dominant Factor Strength by Species



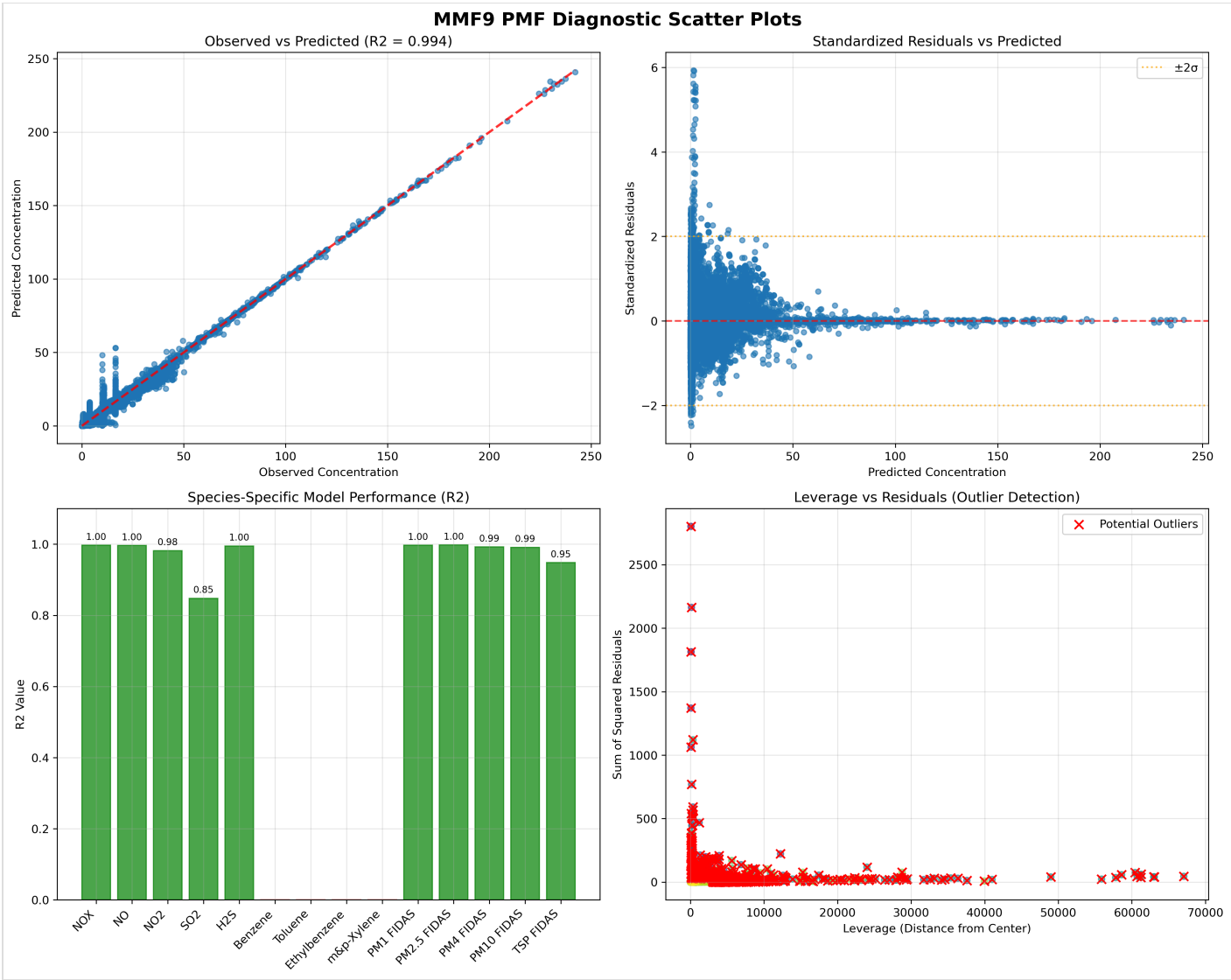
Factor Contribution Variability (CV%)



Mmf9 Mmf 20231001 20231030 Temporal Analysis



Mmf9 Mmf 20231001 20231030 Diagnostic Scatters



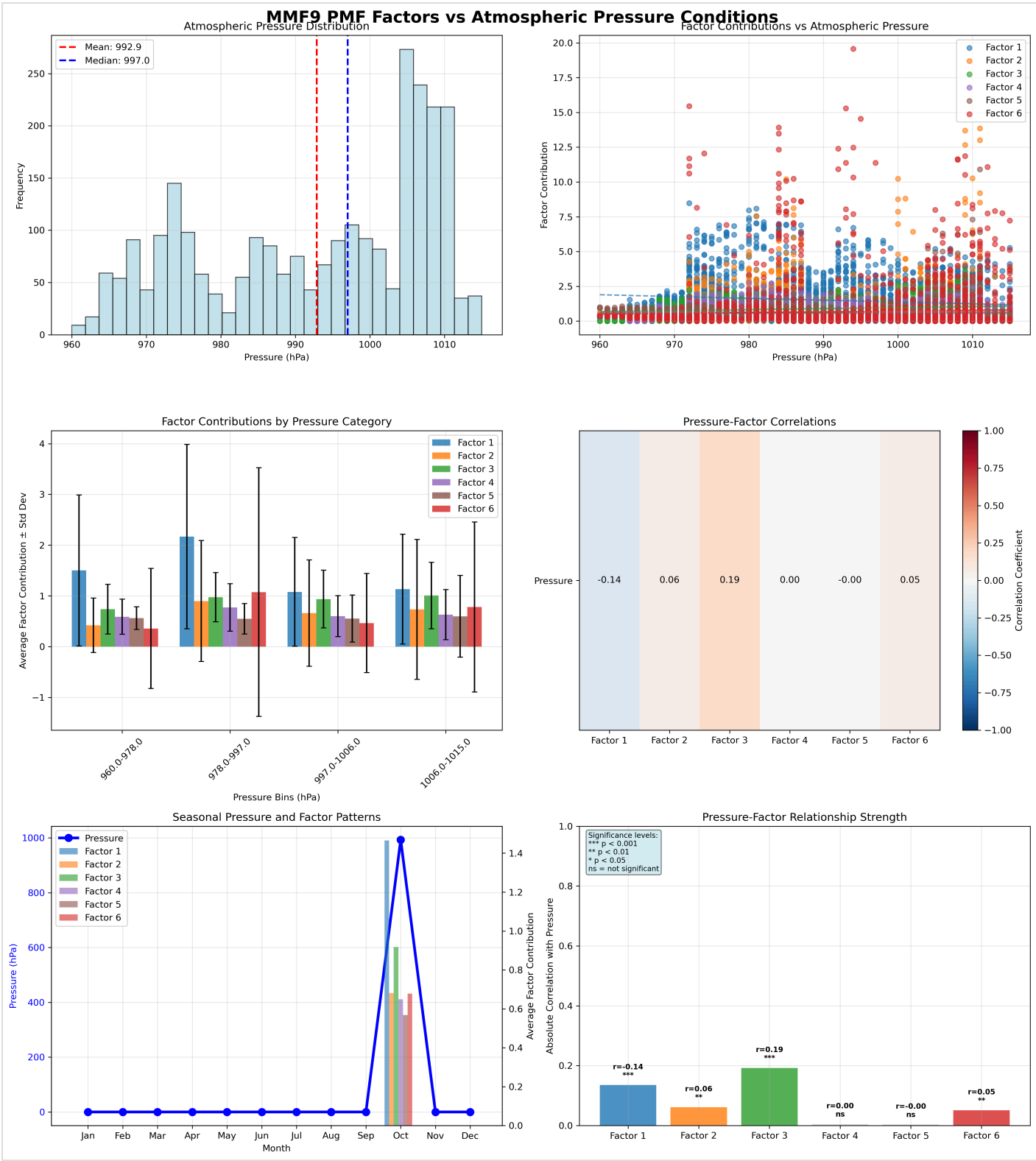
MMf9 Mmf 20231001 20231030 Wind Analysis



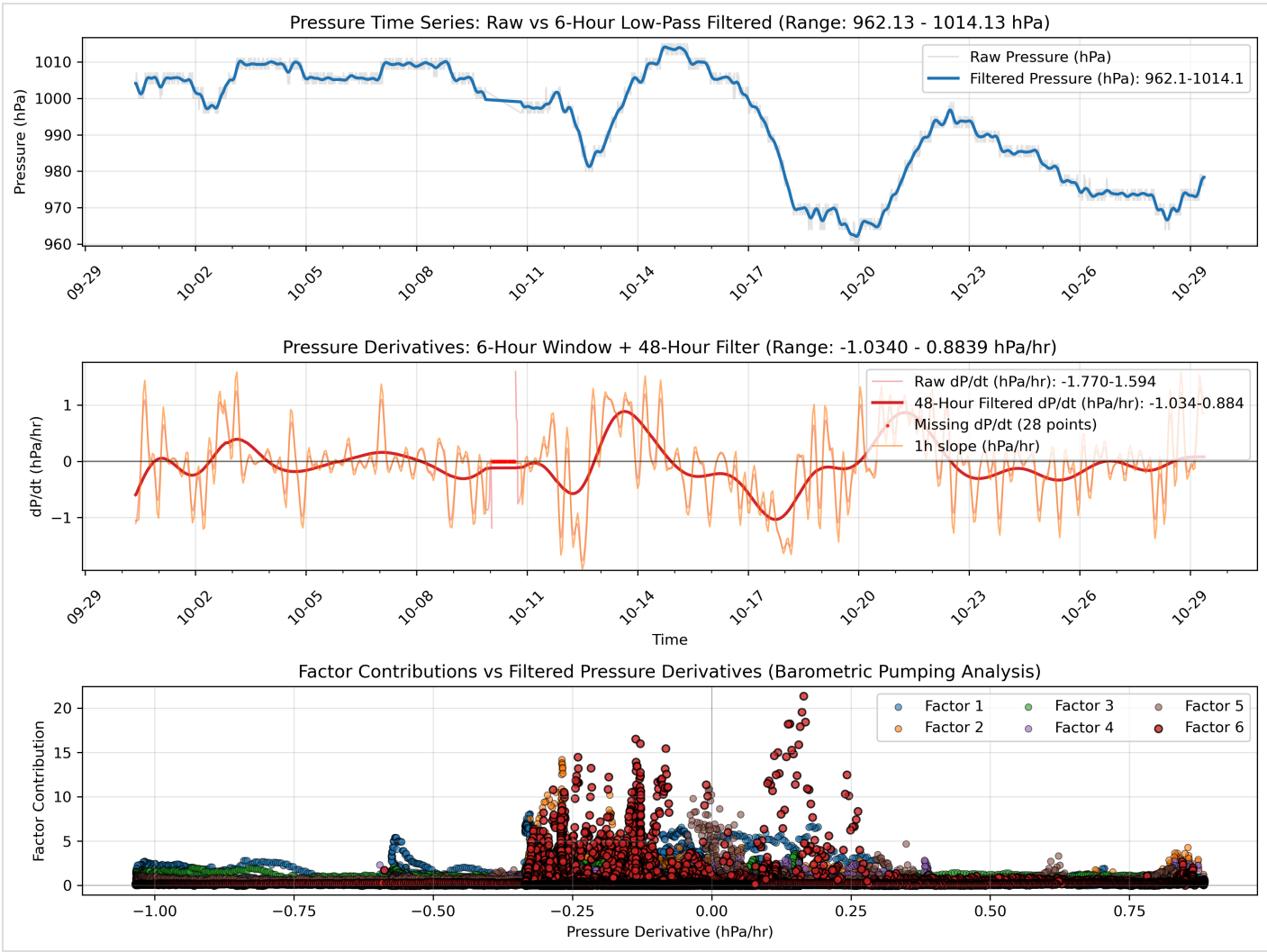
Mmf9 Mmf 20231001 20231030 Temperature Analysis



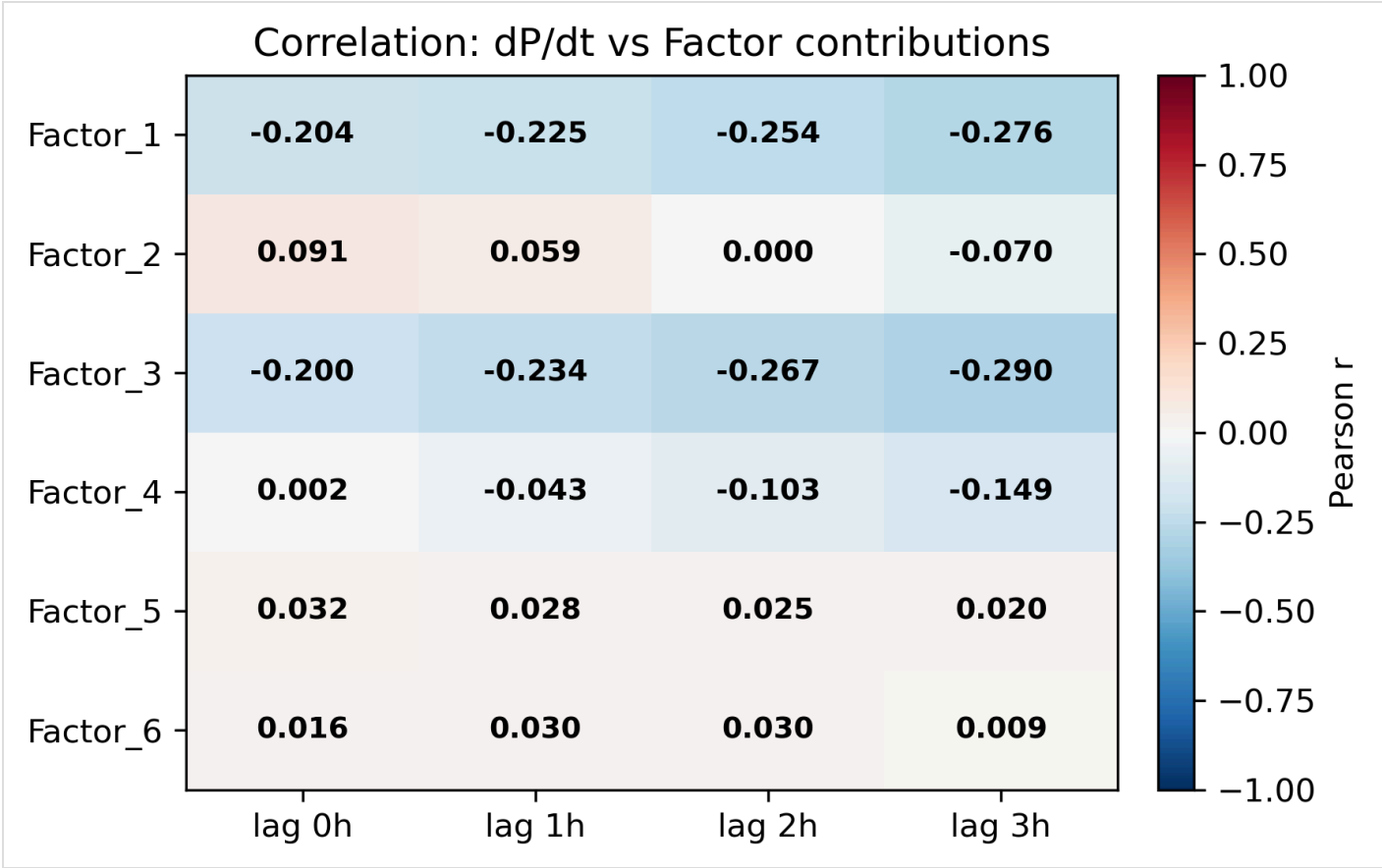
Mmf9 Mmf 20231001 20231030 Pressure Analysis



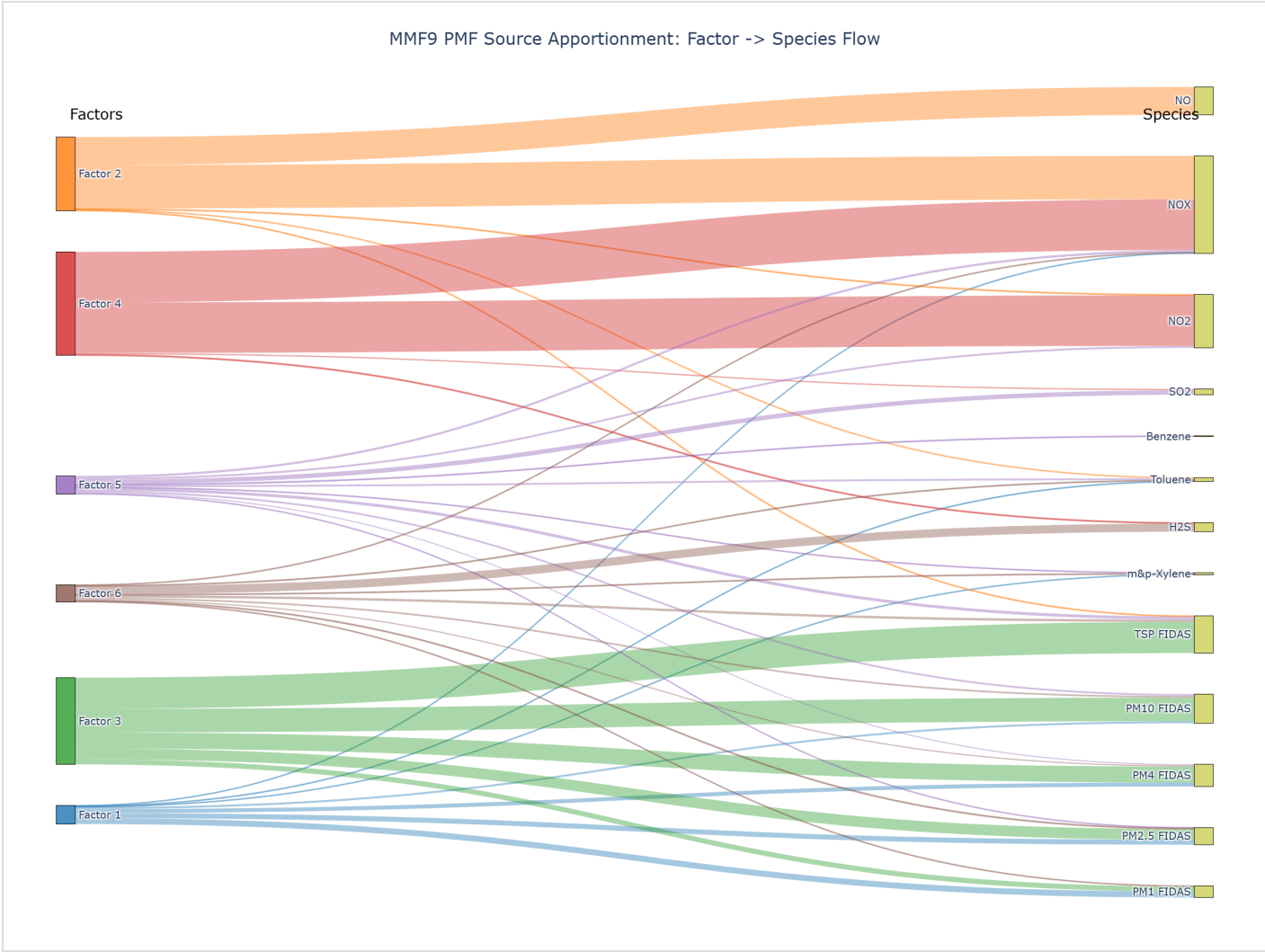
Mmf9 Mmf 20231001 20231030 Pressure Derivative



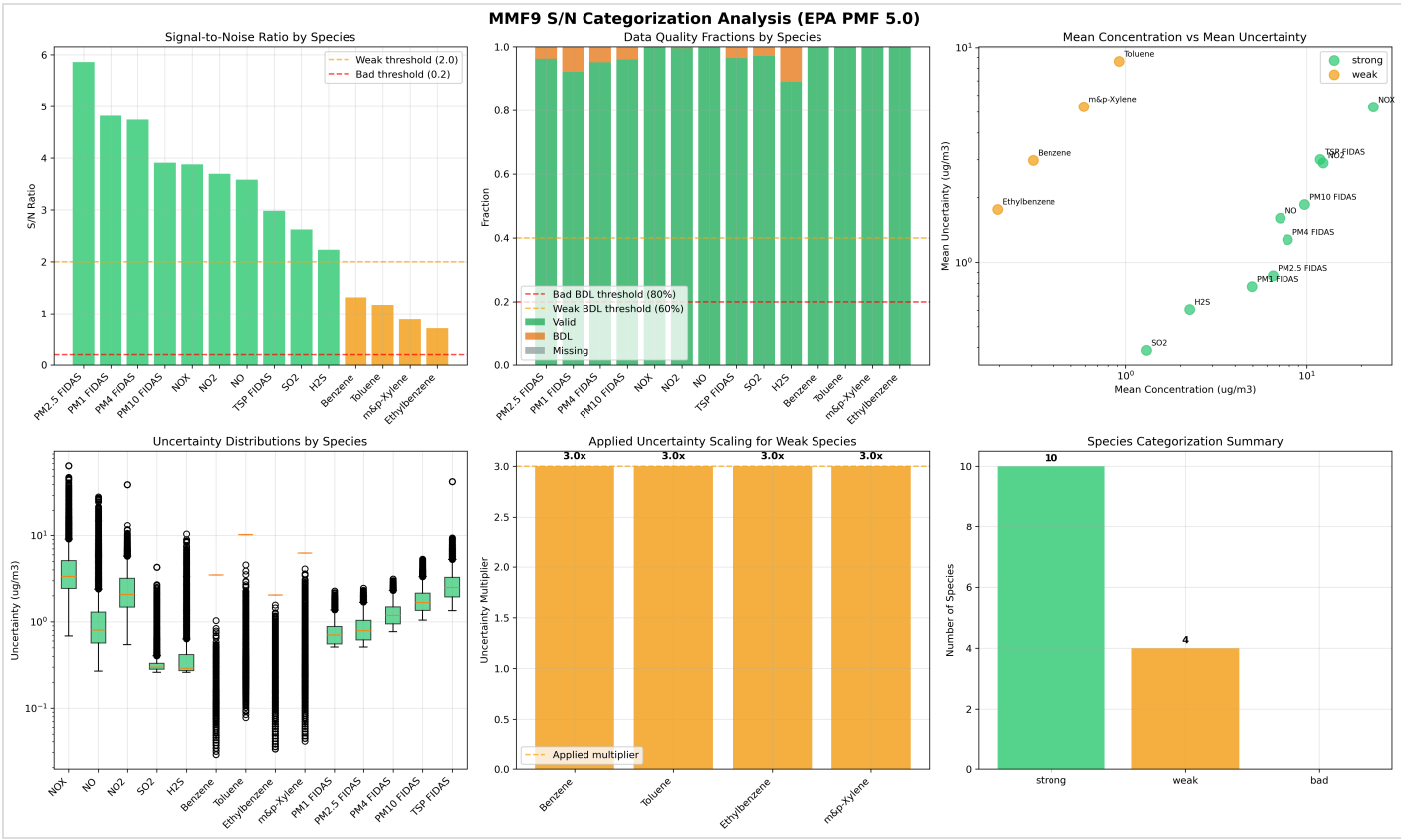
Mmf9 Mmf 20231001 20231030 Dpdt Factor Corr



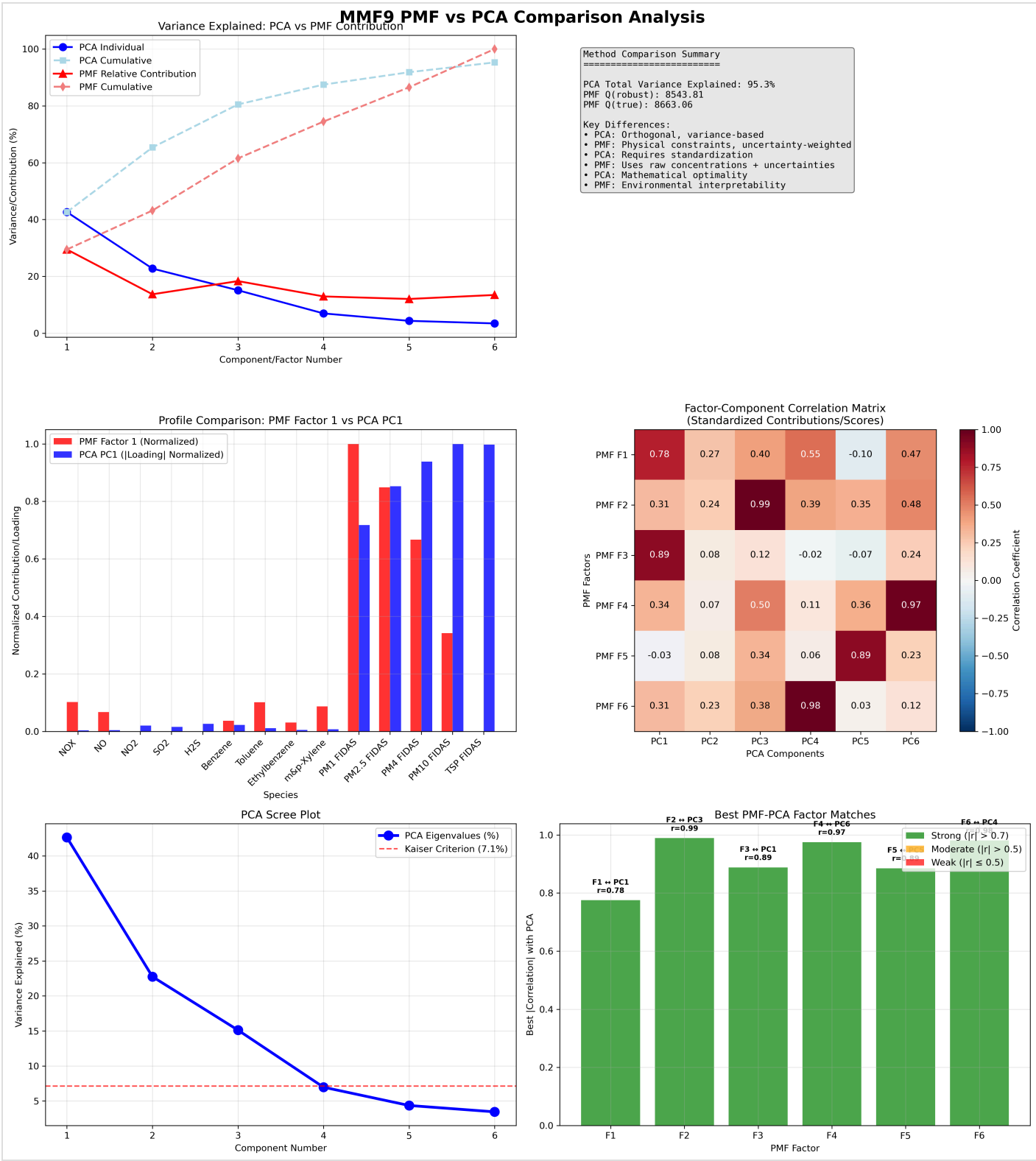
Mmf9 Mmf 20231001 20231030 Sankey Diagram



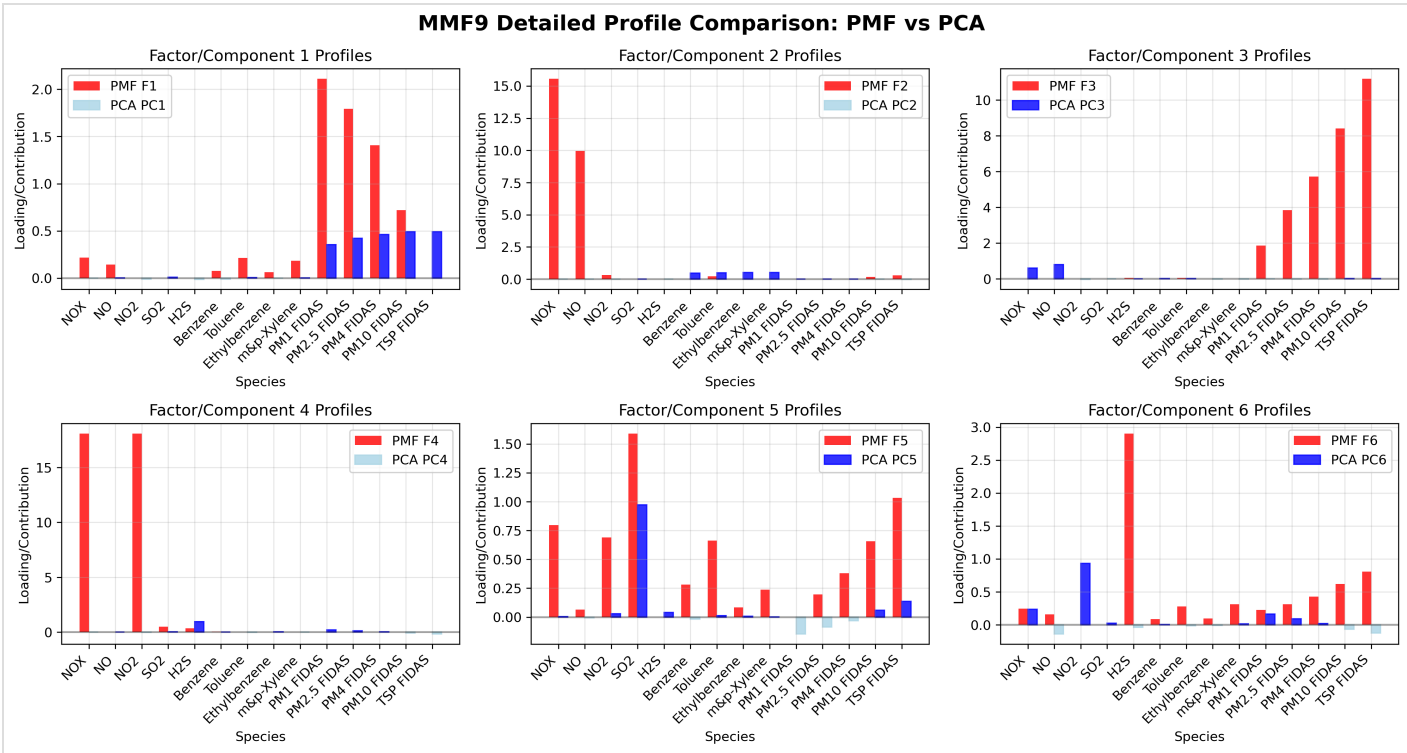
Mmf9 Mmf 20231001 20231030 Snr Analysis



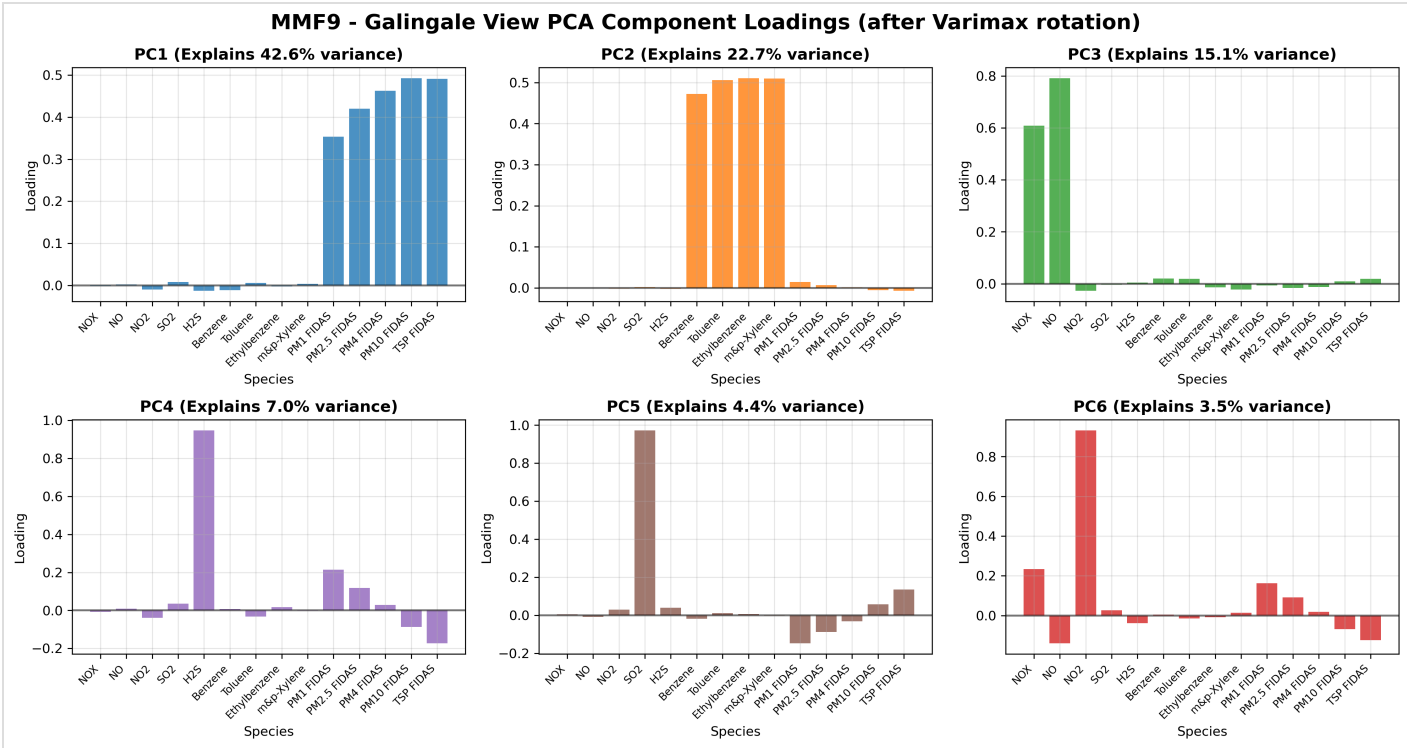
Mmf9 Mmf 20231001 20231030 Pca Pmf Comparison



Mmf9 Mmf 20231001 20231030 Detailed Profile Comparison



Mmf9 Mmf 20231001 20231030 Pca Loadings



[INFO] CLI Flags Used (Reproducibility Record)

Command to reproduce this analysis:

```
python pmf_source_app.py \  
MMF9 \  

```



```
--start-date 2023-10-01 \  
--end-date 2023-10-30 \  
--factors 6 \  
--models 1 \  
--output-dir "check" \  
--uncertainty-mode epa \  
--snr-enable \  
--write-diagnostics \  
--no-weight-aware-init \  
--exclude-species CH4
```

Parameter Details:

Parameter	Value	Description
--station	MMF9	Station identifier (positional argument)
--data-dir	None	Data directory (alternative to station)
--patterns	None	File patterns to match
--start-date	2023-10-01	Analysis start date (YYYY-MM-DD)
--end-date	2023-10-30	Analysis end date (YYYY-MM-DD)
--factors	6	Number of PMF factors
--max-factors	10	Maximum factors to test during optimization
--models	1	Number of PMF models to run
--max-workers	2	Maximum parallel processes
--output-dir	check	Output directory
--run-pca	False	Run PCA analysis for comparison
--create-pdf	False	Create PDF version of dashboard
--remove-voc	False	Remove VOC species from analysis
--scale-units	True	Apply unit standardization to µg/m³
--drop-row-threshold	0.5	Row drop threshold for missing values
--zero-as-bdl	True	Treat zeros as below detection limit
--save-masks	True	Save BDL and missing mask CSVs
--uncertainty-mode	epa	Uncertainty calculation method
--uncertainty-ef-mdl	Built-in values	EF/MDL data source
--uncertainty-epsilon	1e-12	Numerical uncertainty floor
--legacy-min-u	0.1	Minimum uncertainty (legacy mode)

--uncertainty-bdl-policy	five-sixth-mdl	BDL uncertainty policy
--snr-enable	True	EPA S/N-based feature categorization
--snr-weak-threshold	2.0	S/N threshold for weak species
--snr-bad-threshold	0.2	S/N threshold for bad species
--snr-bdl-weak-frac	0.6	BDL fraction for weak categorization
--snr-bdl-bad-frac	0.8	BDL fraction for bad categorization
--snr-missing-weak-frac	0.2	Missing fraction for weak categorization
--snr-missing-bad-frac	0.4	Missing fraction for bad categorization
--exclude-bad	True	Exclude bad species from analysis
--method	ls-nmf	ESAT NMF algorithm (ls-nmf or ws-nmf)
--init-method	column_mean	Matrix initialization method
--init-norm	True	Normalize data before kmeans initialization
--hold-h	False	Hold H (profile) matrix constant during training
--delay-h	-1	Hold H matrix for N iterations (-1 = disabled)
--weight-aware-init	False	Weight-aware initialization for species weighting
--robust-fit	False	Use robust loss during SA training (fallback only)
--robust-alpha	4.0	Robust cutoff alpha for scaled residuals
--dashboard-snr-panel	True	Include S/N panels in dashboard
--write-diagnostics	True	Write diagnostic CSV files
--seed	42	Random seed for reproducibility
--help-detail	False	Show detailed CLI help
--reg-species	None	Species to regularize
--reg-lambda	Default	Regularization strength per species
--reg-template	zero	Template type per species
--reg-template-file	None	Template CSV files
--reg-bursts	5	Number of train->prox cycles
--reg-iter-per-burst	50	Max iterations per burst
--reg-tol	0.0001	Early stop tolerance
--reg-elastic-l1	0.0	Elastic-net L1 penalty

--excluded-species	CH4	Species completely removed from PMF analysis
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